

Comments.

1. Context, problem, and hypotheses

The Thesis addresses a highly relevant problem in autonomous mobile robotics, i.e., reliable simultaneous localization and mapping in dynamic indoor environments using a deliberately restricted sensing configuration, primarily a 2D laser scanner and optionally wheel odometry. The motivation is convincing from an industrial robotics perspective. Conventional SLAM methods are fundamentally easier to formulate when the environment is assumed to be static, whereas real industrial, logistics, retail, and service environments contain people, vehicles, movable equipment, temporary obstacles, and structural reconfigurations. The candidate correctly identifies that such observations can corrupt scan matching, generate blurred or artificially thickened structures in maps, degrade localization, and eventually compromise global map consistency.

A particularly positive aspect is that the thesis does not formulate the problem simply as "dynamic-object detection". Instead, it places dynamics within the broader stability-plasticity problem of long-term mapping. The system must incorporate meaningful environmental changes while avoiding the accumulation of transient observations and redundant information. The distinction between short-lived dynamic objects, semi-static changes, and persistent environmental structure is therefore implicit in the architecture rather than being reduced to a semantic classification problem. This is conceptually appropriate for long-term AMR operation.

Another strength is the deliberate restriction to range sensing. In an era in which many dynamic-SLAM approaches add cameras, 3D LiDARs, semantic networks, or increasingly complex learned perception systems, the thesis asks whether significant robustness can instead be obtained from sensors already commonly present on industrial mobile platforms. The resulting problem is technically harder in some respects because a 2D laser provides little semantic information. At the same time, the restriction makes the potential industrial impact of the work clearer.

The thesis formulates two principal research questions. The first asks how mapping results can be improved when SLAM is performed by a ground mobile robot in a dynamic indoor environment. Four hypotheses are associated with this question: continuously updating local maps using ray tracing, controlling graph growth through relocalization into existing graph representations, removing dynamic observations offline during map generation, and performing offline global map refinement. The second asks how environmental dynamics can be integrated into the localization process. Three corresponding hypotheses investigate geometric features extracted from 2D laser data, distance-map-based registration with robust treatment of outliers, and continuously updated local maps for more stable scan-to-map localization.

The research questions provide a clear conceptual division between mapping quality and localization robustness, while the hypotheses map reasonably well onto the technical developments appearing later in the dissertation. The dissertation does not present a collection of algorithms and attempt to invent a common narrative only at the end; there is a recognizable progression from research problem to algorithmic components and finally to the integrated RicoSLAM framework.

For me, a critical point here concerns the relationship between the two research questions. Localization and mapping in SLAM are inherently coupled, and several mechanisms, particularly local-map updating, appear in hypotheses associated with both questions. The division is understandable organizationally, but scientifically it creates some overlap. A more fundamental overarching hypothesis, followed by specific sub-hypotheses concerning registration, dynamics

removal, graph management, and global consistency, might have provided an even cleaner experimental structure.

2. State of the art, novelty, and impact

Chapter 2 is one of the stronger elements of the work. Rather than providing only a conventional narrative review, the candidate builds the discussion upon a systematic literature review of long-term localization and mapping. The broader review covers 142 works, while the dynamic-environment analysis is subsequently updated and narrowed to 53 studies.

The use of a systematic methodology following PRISMA is valuable because it makes the literature analysis more transparent and reproducible than many robotics thesis literature chapters. The candidate explicitly notes that previous surveys frequently did not describe how papers were selected, limiting reproducibility.

The updated analysis organizes dynamic-SLAM literature according to meaningful dimensions including mapping, localization, environment/domain, sensing configuration, dynamics detection, labeling, statistics, prediction, and removal. This provides a good conceptual basis for positioning RicoSLAM.

The literature review demonstrates broad command of SLAM rather than focusing narrowly on the candidate's own implementation. The progression from classical scan matching, particle-filter SLAM, and pose-graph approaches toward long-term and dynamic-environment SLAM is logical. Another important strength is that the systematic review itself constitutes a scholarly contribution rather than merely background material. This reinforces the maturity of the research program.

The review is broad, but breadth does not automatically establish novelty of every individual algorithmic component. The thesis could distinguish even more sharply between:

1. concepts already established in SLAM;
2. known techniques adapted to the specific 2D/dynamic setting;
3. genuinely new mathematical formulations;
4. new combinations/integration of existing ideas.

This distinction is especially important because distance fields, ray tracing, robust estimators, local maps, graph SLAM, loop closure, and ICP-type registration are individually well-established concepts. The scientific novelty therefore lies substantially in the particular formulations, implementation choices, integration, and experimental demonstration, rather than in originating all underlying principles.

3. Novelty

First, Chapter 3 proposes a line-fitting-based corner-like feature detector in which the keypoint is obtained from the closed-form intersection of locally fitted lines at multiple distance-based neighborhood scales. An attractive property is that the feature position does not need to correspond directly to an actual laser return. The thesis reports encouraging repeatability, including 91.5% of associations among the 39 most persistent features having standard deviation at or below 0.03 m.

Second, Chapter 4 develops the distance-map-based registration formulation. Instead of performing explicit nearest-neighbor searches as in conventional ICP, the method accesses a precomputed distance representation, potentially allowing lookup up to $O(1)$. Both point-to-point and point-to-plane formulations are derived under Gauss-Newton optimization on $SE(2)$, including multiple representation levels and dense/sparse alternatives.

This is one of the technically strongest portions of the thesis because the contribution is not merely implementation-oriented. There is mathematical formulation and derivation behind the registration procedure.

Third, and arguably most importantly, the candidate integrates these ideas into RicoSLAM, combining a distance-map-based front end with a factor-graph back end, keyframe management based on inlier ratios, loop detection, relocalization into existing graph regions, online local-map maintenance, offline dynamics removal, and distance-map-based GBA factors.

The potential impact is strongest in industrial AMRs operating in planar environments. The thesis demonstrates that a relatively modest sensing arrangement can produce strong localization and mapping performance without relying on semantic perception or high-dimensional sensors.

Another positive contribution to impact and reproducibility is the availability of public datasets/software and the associated peer-reviewed publications. The dissertation includes contributions related to the IILABS 3D dataset and reports application of RicoSLAM in a container-unloading scenario.

Important limitation concerning novelty/impact is that the complete conceptual architecture is not yet validated end-to-end. Most notably, the distance-map-based GBA component is tested separately on a synthetic Manhattan-like benchmark rather than being fully integrated and evaluated in the complete online RicoSLAM pipeline. The thesis itself explicitly acknowledges that its integration and end-to-end real-world validation remain future work.

This matters because the abstract and architecture may initially give the impression that GBA constitutes a fully validated final component of RicoSLAM. Scientifically, however, one should distinguish between a validated mathematical GBA formulation and a validated end-to-end RicoSLAM+GBA system. The dissertation demonstrates the former much more strongly than the latter.

3. Methodology, analysis, and validation

The thesis contains substantial experimental work at multiple levels rather than relying on a single final benchmark. The corner detector is evaluated for observability and association consistency. The registration framework is evaluated first synthetically and subsequently through real-world laser odometry. RicoSLAM is then evaluated using trajectory accuracy, graph scalability, dynamic-removal behavior, and global optimization experiments.

The trajectory analysis also employs standard quantitative metrics. ATE measures global trajectory consistency, while RTE and RRE characterize local translational and rotational behavior over fixed 10 m trajectory segments. The main SLAM comparison includes GMapping and SLAM Toolbox under multiple configurations rather than selecting only one arbitrary baseline configuration. This is good experimental practice.

Another strong methodological aspect is that the thesis does not restrict "scalability" to trajectory accuracy. It explicitly evaluates the number of keyframe nodes, incorporated scans, and graph edges, which is directly relevant to long-term operation.

3.1. Limited SLAM baseline diversity

The full system is compared primarily with GMapping and SLAM Toolbox. These are legitimate and widely recognized 2D SLAM references, but they do not exhaust the relevant state of the art. The thesis itself acknowledges that comparison with additional systems such as Cartographer remains future work. **For a thesis claiming advancement of the state of the art, this is perhaps the clearest experimental weakness. A broader benchmark would make the superiority claims much stronger.**

- **QUESTION:** Well, why were only GMapping and SLAM Toolbox selected for the principal system-level comparison, and how confident are you that the conclusions would remain unchanged against a broader set of modern 2D SLAM systems?

3.2. Computational efficiency

A central motivation of distance maps is the replacement of explicit nearest-neighbor searches with potentially $O(1)$ lookup. This is theoretically attractive.

However, the final RicoSLAM experiments apparently do not provide a comprehensive end-to-end timing and memory benchmark separating front-end computation, back-end optimization, loop closure, dynamics removal, and map storage. Indeed, explicit measurement of front-end/back-end computation times and memory consumption is identified as future work.

3.3. GBA validation is substantially weaker than the rest of the thesis

The proposed GBA factors are mathematically interesting, but the experimental validation is synthetic and isolated. The dissertation openly acknowledges that full integration into RicoSLAM and end-to-end real-world validation remain unfinished.

Thus, for me, this means that the hypothesis 1.4 is only partially supported compared with H1.1 to H1.3 and the localization hypotheses.

- **QUESTION:** So, do you consider Hypothesis 1.4 fully validated, or only demonstrated in principle? What evidence would still be required before claiming that the GBA stage improves RicoSLAM under real dynamic operating conditions?

3.4. Dynamic-map quality lacks conventional external ground truth

The occupancy-distribution analysis is useful and the visual maps appear cleaner after removal of transient returns. However, "cleaner" occupancy probabilities do not necessarily prove that every removed point was truly dynamic.

Ideally, one would like a manually labeled or independently measured static ground-truth map allowing precision/recall, false-removal rate, missed-dynamic rate, structural-map error, or similar measures.

This is especially important because an aggressive filtering mechanism can improve apparent map sharpness while accidentally removing legitimate but infrequently observed static structure.

This does not invalidate the experiment, but it limits the strength of the interpretation.

4. Results

The thesis contains many quantitative comparisons, but the validation could be stronger with greater attention to statistical uncertainty and repeatability.

SLAM pipelines can depend strongly on parameter choices, initial conditions, sensor synchronization, and particular trajectory geometry. Where algorithms are deterministic, repeated trials are less critical, but uncertainty can still enter through preprocessing and experimental acquisition.

Confidence intervals, parameter sensitivity analyses, or systematic robustness studies would provide stronger evidence than best-configuration comparisons alone.

This is especially relevant because Table 5.1 compares several configurations and identifies best-performing variants. Whenever a "best configuration" is selected after experimentation, one should ask how robust the conclusion is to parameter variation.

5. Overall quality of the dissertation

In my view, the strongest quality of the thesis is not any single algorithm but the progression from theoretical and algorithmic developments to an integrated SLAM framework.

The document has a clear organization of the chapters.

The work also demonstrates considerable technical breadth. It spans geometric perception, feature extraction, point-cloud registration, nonlinear least-squares optimization on $SE(2)$, robust estimation, distance fields, pose graphs, loop closure, relocalization, occupancy mapping, dynamic-scene filtering, and bundle adjustment. That breadth is appropriate for doctoral-level robotics research.

Another strong point is the combination of theoretical and applied work. The candidate derives optimization formulations rather than merely combining software packages, but also implements and evaluates an actual SLAM system. The use of real robotic datasets and industrially relevant scenarios strengthens the engineering contribution.

The main weakness is an imbalance in validation maturity across the contributions. The feature detector and registration components receive detailed component-level study. RicoSLAM receives real-world system-level evaluation. In contrast, GBA remains essentially a proof-of-concept validated synthetically. Consequently, the final architecture is slightly ahead of its experimental validation.

Second, the baseline comparison at system level is narrower than desirable. RicoSLAM clearly performs very well against GMapping and SLAM Toolbox on the evaluated sequences, but this supports the precise conclusion that RicoSLAM outperforms those baselines under those experiments. It is not sufficient by itself to establish universal superiority over contemporary 2D SLAM.

Thus, I would distinguish carefully between the parts of RicoSLAM that are convincingly demonstrated experimentally and those that remain promising but incompletely validated. The trajectory results against GMapping and SLAM Toolbox are strong, as are the graph-size results and the evidence that the dynamics-removal procedure improves occupancy-map consistency. However, the absence of broader state-of-the-art comparisons, explicit system-level timing and memory evaluation, independently labeled dynamic-map ground truth, and especially real-world

end-to-end GBA validation limits how broadly some of the conclusions can be generalized. The candidate himself identifies several of these issues as future work, which demonstrates appropriate awareness of the boundaries of the current contribution.